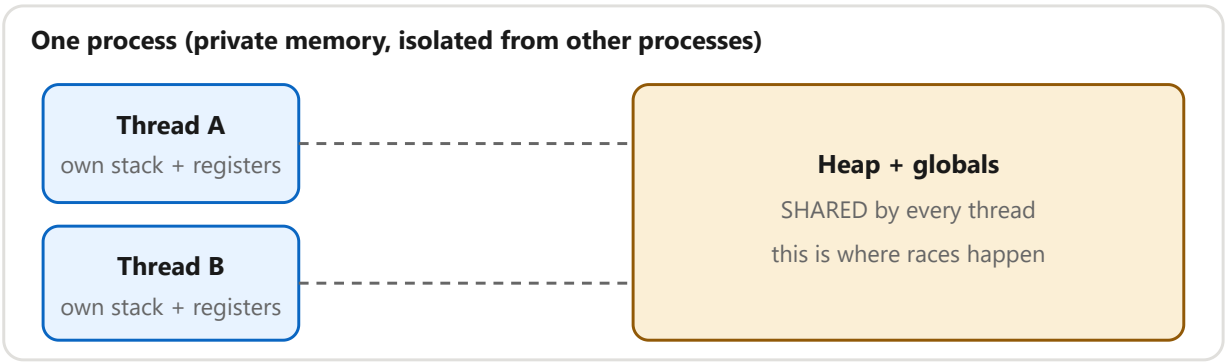
Multithreading

From your first thread to data races, mutexes, deadlocks and condition variables: the model, the rules and the bugs. Atomics, futures and pools continue in Tasks and Parallelism and Atomics and the Memory Model.

Why threads? The mental model



Threads share memory; that sharing is the source of both the speed and the bugs.

A **thread** is an independent flow of execution. Several threads make progress at once, like several cooks in one kitchen.

TERM	MEANING
Concurrency	tasks *in progress* at overlapping times (can be interleaved on one core)
Parallelism	tasks running at the *same instant* on different cores

Why: speed (use all cores), responsiveness (keep a UI or network handler alive), hiding I/O waits. **Cost:** about 1 MB stack each, startup time, context switches. Use threads only where the gain is measurable.

> THE WHOLE DIFFICULTY Threads share memory, and the OS can switch between them **between any two instructions**.

```cpp
std::thread::hardware_concurrency();     // how many hardware threads (cores x hyperthreads) this machine has, e.g. 8
```

Your first thread

```cpp
#include <iostream>
#include <thread>

void hello(int id) {
    std::cout << "worker " << id << " running\n";
}

int main() {
    std::thread t(hello, 1);              // starts running hello(1) IMMEDIATELY, on a new thread
    std::cout << "main continues\n";      // main keeps going at the same time
    t.join();                             // wait here until t has finished
}
```

The **order is not fixed**. Run it repeatedly and you may see either:

```
main continues            worker 1 running
worker 1 running    or    main continues
```

A test that passed once proves almost nothing.

Rules

- **Always `join()` or `detach()` before the `std::thread` is destroyed**, otherwise `std::terminate`. A detached thread must not touch anything that may be destroyed.
- **Arguments are copied** into the thread. Use `std::ref` to pass a reference.
- Threads are **move-only**: `std::thread b = std::move(a);`.
- An exception escaping the thread function calls `std::terminate`. Catch inside, or pass it out with a `future`.

```cpp
void addOne(int& n) { ++n; }

int x = 0;
std::thread t1(addOne, std::ref(x));      // std::ref: pass x by reference
(without it this will not even compile)
t1.join();                                // x is now 1

std::thread t2([] { std::cout << "a lambda works too\n"; });
t2.join();

struct Worker { void run(int n) { /* ... */ } } w;
std::thread t3(&Worker::run, &w, 5);      // member function: pointer to
member, the object, then the arguments
t3.join();

std::this_thread::sleep_for(std::chrono::milliseconds(100));   // pause the
current thread
std::this_thread::get_id();                                    // identify
the current thread
```

CLASSIC DETACH BUG

```cpp
void f() {
    int local = 0;
    std::thread t([&] { ++local; });   // captures local BY REFERENCE
    t.detach();
}                                      // f returns, local is
destroyed, the thread still writes to it
```

Capture by value (or `join()`) if the thread can outlive the variable.

CONVERTED TOO LATE

```
void f(int i, const std::string& s);

void oops() {
    char buffer[1024];
    std::thread t(f, 3, buffer);               // copies the char*
POINTER, not the text
    t.detach();                                // oops() may return
before the thread builds the std::string
}                                              // fix: std::thread t(f,
3, std::string(buffer));
```

The conversion to `std::string` happens later, inside the new thread. Convert on the calling side.

Join on every path

If anything before `join()` can throw, the thread stays joinable and the program terminates. Use RAII:

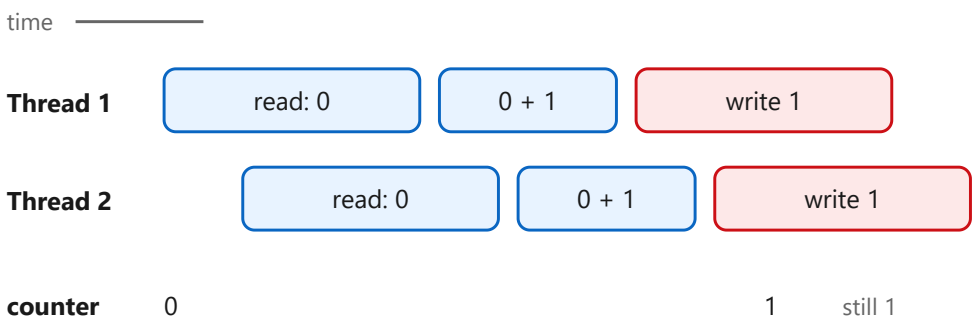
```
class ThreadGuard {
    std::thread& t_;
public:
    explicit ThreadGuard(std::thread& t) : t_(t) {}
    ~ThreadGuard() { if (t_.joinable()) t_.join(); }     // joins however the
scope is left
    ThreadGuard(const ThreadGuard&) = delete;
    ThreadGuard& operator=(const ThreadGuard&) = delete;
};

void f() {
    std::thread t(work);
    ThreadGuard g(t);                  // joins when f exits, even by an
exception
    mayThrow();
}
```

Why no implicit join or detach? A join hides long waits; a detach leaves a thread using dead locals. If a class owns a `std::thread`, declare it **last** among the members.

> **C++20: STD::JTHREAD** Joins automatically in its destructor and supports `std::stop_token` cancellation.

Shared data and data races



Expected 2, got 1: both threads read 0 before either wrote. One update is lost.

A data race: the two read-modify-write sequences overlap. ++counter is three steps, not one.

Two threads each increment a shared counter:

```
int counter = 0;
void work() { for (int i = 0; i < 2; ++i) ++counter; }   // two threads each run this: expect counter == 4
```

`++counter` is **three** CPU steps:

1. **load** the value from memory into a register: `r = counter`
2. **add** one to the register: `r = r + 1`
3. **store** the register back to memory: `counter = r`

If B loads between A's load and store, **one increment is lost**.

STEP	THREAD	WHAT RUNS	R IN THAT THREAD	COUNTER AFTERWARDS
1	A	`r = counter`	0	0
2	B	first increment: `r = counter`, `r = r + 1`, `counter = r`	1	1
3	B	second increment, the same three lines	2	2
4	A	`r = r + 1` (still working from the old value 0)	1	2
5	A	`counter = r` overwrites both of B's increments	1	1
6	A	second increment: `r = counter`, `r = r + 1`, `counter = r`	2	2

One bad schedule: the final value is **2**, not 4. Two increments were lost.

> **CAVEAT** A real race may fail once in thousands of runs, which is why it survives testing.

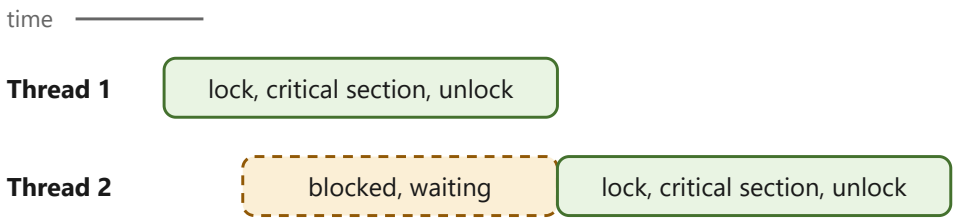
> **DATA RACE = UNDEFINED BEHAVIOR** Two threads access the same memory, at least one writes, no synchronization. A **race condition** is the wider timing-dependent logic bug.

Why it hurts: shared data has an **invariant** (links agree, count matches items). An update breaks it briefly, and a thread looking then sees corruption. Trouble starts when *two or more* pieces of data change together.

Three ways to be safe

STRATEGY	HOW
1. Do not share	own data per thread (locals, `thread_local`), merge afterwards. Best option.
2. Share, never modify	`const` data is safe to read from many threads.
3. Share and protect	a **mutex** or an **atomic**.

Mutexes: taking turns



Only one thread is inside the protected region at any moment, so the updates cannot overlap.

Cost: thread 2 does nothing while it waits. Keep critical sections short.

A **mutex** lets one thread at a time into the **critical section**; others **wait** at `lock()`. Same two threads, now locked.

With the lock, the result is **always 4**.

```
std::mutex m;
int counter = 0;

void work() {
    for (int i = 0; i < 1000; ++i) {
        std::lock_guard<std::mutex> lock(m);   // locks now, unlocks
automatically at the end of the block
        ++counter;
    }
}
```

Use RAII locks, never raw lock()/unlock()

An exception or early return between `lock()` and `unlock()` leaves the mutex locked forever.

TYPE	USE IT WHEN
`std::lock_guard<std::mutex>`	default: lock for the whole scope
`std::unique_lock<std::mutex>`	unlock early, relock, defer; required by `cv.wait`
`std::scoped_lock` (C++17)	several mutexes at once, deadlock-free
`std::shared_mutex` + `shared_lock`	many readers or one writer

Protecting a class

```
class SafeCounter {
    mutable std::mutex m_;        // mutable: so const getters can still lock
it
    int value_ = 0;
public:
    void increment()   { std::lock_guard<std::mutex> lk(m_); ++value_; }
    int  get() const   { std::lock_guard<std::mutex> lk(m_); return value_; }
};
```

GOOD LOCKING

- Keep critical sections **short**; do slow work (I/O, computation) outside.
- Mutex **next to its data**; never hand out references to that data.
- Never call unknown code (callbacks, virtuals) while locked.
- Too much locking means **contention**: the program becomes serial.

Other mutex types and `call_once`

TYPE	WHAT IT ADDS	USE WHEN
`std::mutex`	nothing: one owner, not re-lockable	the default
`std::recursive_mutex`	the **same thread** may lock it again; it must unlock as many times	almost never: see below
`std::timed_mutex`	`try_lock_for(d)` and `try_lock_until(t)`: give up after a time	a thread must not wait forever (watchdogs, real-time deadlines)
`std::shared_mutex` (C++17)	**shared** mode for readers, **exclusive** mode for writers	reads dominate and the critical section is not tiny

recursive_mutex is a design smell

```
class Account {
    std::recursive_mutex m_;
    int balance_ = 0;
public:
    void deposit(int n)      { std::lock_guard lk(m_); balance_ += n; }
    void depositTwice(int n) { std::lock_guard lk(m_); deposit(n);
deposit(n); }   // locks the SAME mutex again
};
```

It works (compiled: balance 10), but it hides a design problem: `depositTwice` does not know what state `deposit` assumes. Prefer a **public** method that locks once and a **private** `depositLocked()` that assumes the lock is held. A plain `std::mutex` locked twice by one thread is undefined behaviour (usually a deadlock).

Reader-writer lock

```
class PriceTable {
    mutable std::shared_mutex m_;
    std::map<std::string, double> px_;
public:
    void set(const std::string& k, double v) { std::unique_lock lk(m_);
px_[k] = v; }   // exclusive: one writer, no readers
    double get(const std::string& k) const {
        std::shared_lock lk(m_);
// shared: any number of readers together
        auto it = px_.find(k); return it == px_.end() ? -1 : it->second;
    }
};
```

- A `shared_mutex` costs more than a `mutex` per lock. It pays off only when there are **many readers** and each read holds the lock for a while.
- Writers can starve if readers never leave (implementation-defined fairness).
- `try_lock()` returns immediately with true or false: compiled, a second thread's `try_lock` on a held mutex returned false; `try_lock_for(20 ms)` on one held for 200 ms returned false after 20 ms.

`std::call_once`

```
std::once_flag flag;
void init() { std::call_once(flag, [] { loadConfig(); }); }   // eight
threads may call init(): loadConfig runs exactly once
```

Use it (or a function-local `static`) for lazy initialization shared by several threads, instead of a hand-written double-checked lock.

> KEY POINT "When would you use a recursive mutex?" Rarely, and say so: it usually means a method that locks calls another that also locks. Restructure into a locking public method and a lock-assuming private one.

Races hidden in interfaces

A mutex in every member function is necessary but **not sufficient**. Two threads run this on a shared stack whose methods all lock internally:

```
SafeStack<int> s;                  // shared by several threads; push, pop,
top, empty all lock internally

if (!s.empty()) {                  // 1. is there anything to take?
    int v = s.top();               // 2. read the top element
    s.pop();                       // 3. remove it
    use(v);
}
```

Each call is atomic, the **sequence is not**.

Two failures, neither preventable by internal locking:

- B calls `top()` on an empty stack: undefined behavior (the `empty()` answer went stale).
- Both read `top()` before either pops: one element processed **twice**.

This **check-then-act** race also hits lock-free structures. The cause is the interface, so change the interface.

Fix: offer whole operations

```
template <typename T>
class SafeStack {
    std::stack<T> s_;
    mutable std::mutex m_;
public:
    void push(T v) {
        std::lock_guard<std::mutex> lk(m_);
        s_.push(std::move(v));
    }
    std::optional<T> try_pop() {                // check + read + remove as
ONE atomic operation
        std::lock_guard<std::mutex> lk(m_);
        if (s_.empty()) return std::nullopt;
        T v = std::move(s_.top());
        s_.pop();
        return v;
    }
    // empty() may still exist, but treat its answer as a hint only: it can
be stale the moment it returns.
};

if (auto v = s.try_pop()) use(*v);              // no window between the
check and the removal
```

WHY DOES STD::STACK SPLIT TOP() AND POP()? If `pop()` returned by value and the copy threw after removal, the element would be lost. A concurrent version must merge them: out-parameter, `std::optional`, `shared_ptr<T>`, or a non-throwing move.

Same pattern elsewhere:

RACY SEQUENCE	WHOLE-OPERATION FIX
`if (!m.contains(k)) m[k] = v;`	one `insert` / `try_emplace` call that does both under the lock
`if (q.size() < max) q.push(x);`	a `try_push` that checks and pushes under one lock
`if (!file_exists(p)) create(p);`	an atomic "create if absent" call (for example the OS `O_EXCL` flag)
`if (counter.load() > 0) counter--;`	a `compare_exchange` loop

KEY POINT "Mutex in every method, so thread-safe?" No: each call is safe, the caller's *sequence* is not. Provide combined operations like `try_pop` and protect the **whole invariant**.

Deadlock

A **deadlock**: threads wait for each other forever. A holds `a` wants `b`; B holds `b` wants `a`.

STEP	THREAD	RUNS	STATE AFTERWARDS
1	A	`a.lock()`	A holds a
2	B	`b.lock()`	B holds b
3	A	`b.lock()`	A waits for b, which B holds
4	B	`a.lock()`	B waits for a, which A holds: **deadlock**, neither can ever continue

In real programs it may occur once in a million runs, only in production.

It needs all four (Coffman conditions); remove one and it cannot happen:

1. **Mutual exclusion**: one holder at a time
2. **Hold and wait**: holds one lock, waits for another
3. **No preemption**: a lock cannot be taken away
4. **Circular wait**: A waits for B, B for A

Fix: one global lock order

If **every** thread takes `a` before `b`, no cycle is possible.

```
std::mutex a, b;

void thread1() { std::scoped_lock lk(a, b); /* ... */ }   // C++17: locks
both with a deadlock-avoidance algorithm
void thread2() { std::scoped_lock lk(b, a); /* ... */ }   // OK even though
the order differs: scoped_lock sorts it out

// pre-C++17:
std::lock(a, b);                                          // lock both
atomically (deadlock-free)
std::lock_guard<std::mutex> la(a, std::adopt_lock);       // adopt the
already-held locks so RAII still unlocks
std::lock_guard<std::mutex> lb(b, std::adopt_lock);
```

Four guidelines, one idea

Do not wait for a thread that might be waiting for you.

1. **Avoid nested locks.** If you need several, take them in one step with `std::scoped_lock`.
2. **No user-supplied code under a lock.** A callback may take another lock, or keep a pointer to your data:

```
class Wrapper {
    Data data_;
    std::mutex m_;
public:
    template <typename F>
    void process(F f) {
        std::lock_guard<std::mutex> lk(m_);
        f(data_);                          // hands a reference to the
protected data to unknown code
    }
};

Data* leaked;
wrapper.process([](Data& d) { leaked = &d; });    // the callback keeps the
pointer
leaked->modify();                                  // later: modifies the
data with NO lock held
```

   Never pass out a pointer or reference to protected data.
3. **Fixed lock order.** Hand-over-hand locking on list nodes deadlocks when two threads walk in opposite directions. Fix the traversal direction.
4. **Lock hierarchy.** Each mutex has a level; a thread may only lock a *lower* level than it holds. Violations throw the first time, even if the unlucky timing never occurs. It is a small wrapper around `std::mutex` that records the current level in a `thread_local` variable and throws if you lock upward. The idea is worth knowing; in practice, `std::scoped_lock` and a documented lock order cover most programs.

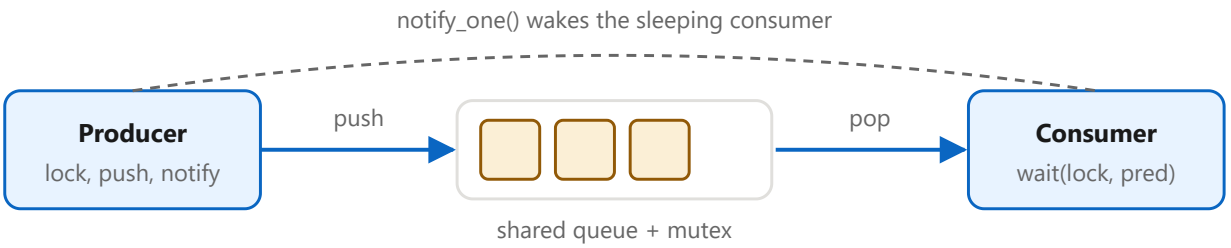
NO MUTEX NEEDED Two threads that `join()` each other deadlock, as does any cycle of futures or condition-variable waits.

```
std::thread tA, tB;                    // both threads are started
elsewhere, running funcA and funcB

void funcA() { tB.join(); }            // A waits for B to finish ...
void funcB() { tA.join(); }            //  ... and B waits for A to finish:
neither ever returns
```

KEY POINT "Prevent deadlock?" Global lock order; `std::scoped_lock`; no foreign code under a lock; short locks; no nesting; `try_lock` with back-off. Re-locking a non-recursive mutex you own is undefined behavior.

Condition variables



The consumer sleeps (no CPU) until notified, then re-checks the predicate to survive spurious wakeups.

Condition variable: sleep until something changed, always paired with a mutex and a condition.

To **wait until something happens**, a busy loop burns 100% CPU. A **condition variable** lets the thread *sleep* until woken, always together with a mutex and the shared state.

```
std::mutex m;
std::condition_variable cv;
std::queue<int> q;

// CONSUMER
std::unique_lock<std::mutex> lk(m);
cv.wait(lk, [] { return !q.empty(); });     // sleep until the predicate is
true
int x = q.front(); q.pop();

// PRODUCER
{ std::lock_guard<std::mutex> lk(m); q.push(42); }
cv.notify_one();                             // wake one waiting thread
```

Three rules to follow

1. **Always wait with a predicate.** Spurious wakeups happen, and another consumer may take the item first.
2. **Change the state under the mutex**, then notify. Otherwise a lost wakeup: the waiter sees false, you notify, it then sleeps forever.
3. The waiter needs `unique_lock`: `wait` unlocks while sleeping.

`notify_one()` wakes one waiter; `notify_all()` wakes all (shutdown, or several may proceed).

A reusable thread-safe queue

```
template <typename T>
class SafeQueue {
    std::queue<T> q_;
    std::mutex m_;
    std::condition_variable cv_;
public:
    void push(T v) {
        { std::lock_guard<std::mutex> lk(m_); q_.push(std::move(v)); }
        cv_.notify_one();                       // notify AFTER releasing the
lock: the woken thread can take it at once
    }
    T pop() {                                   // blocks until an item exists
        std::unique_lock<std::mutex> lk(m_);
        cv_.wait(lk, [this] { return !q_.empty(); });
        T v = std::move(q_.front());
        q_.pop();
        return v;
    }
};
```

Debugging and common bugs

Timing bugs vanish under a debugger or a `printf` ("Heisenbug"). Use tools:

- **ThreadSanitizer**: `-fsanitize=thread -g` finds data races and prints both accesses. Run tests with it.
- **Helgrind / DRD** (Valgrind): races and lock-order violations.
- Thread ids in logs; `gdb` `thread apply all bt` on a hang.

SYMPTOM	LIKELY CAUSE
Wrong totals, rare crashes	data race
Hang, CPU idle	deadlock or lost wakeup
Hang, CPU 100%	busy-wait or livelock
Crash at exit	detached thread using destroyed data; joinable thread destroyed
Slower with more threads	lock contention, false sharing, oversubscription
Garbled output	several threads on `std::cout`: build the line, print once, or lock

Vocabulary

- **Thread-safe**: usable from several threads without external locking (`shared_ptr`'s reference count, *not* its pointee).
- `thread_local`: one copy per thread.
- **Function-local** `static` init is thread-safe since C++11 (Meyers Singleton).
- `std::call_once(flag, fn)`: initialize exactly once.

> WHERE NEXT Atomics and the memory model are in Atomics and the Memory Model, futures, thread pools and C++20 coordination tools in Tasks and Parallelism, and false sharing, contention and pinning in Performance & Memory.

Further reading: your notes in Module 3: multithreading and synchronization (with runnable examples), the modern-cpp-tutorial concurrency chapter (threads, mutexes, futures, condition variables, atomics and the memory model), and the modern-cpp-features notes on `std::thread`, `std::async`, `std::scoped_lock` and parallel algorithms.

Further reading by section

Where each section of this lesson is explained in more depth. Pages are PDF viewer pages (the number your PDF reader shows).

SECTION	BOOK	PART OF THE BOOK	PDF PAGES
Why threads? The mental model	C++ Concurrency in Action	1.1 What is concurrency?	25 to 30
	C++ Concurrency in Action	1.2.3 When not to use concurrency	32 to 33
Your first thread	Effective Modern C++	Item 37, Make std::threads unjoinable on all paths	268 to 276
	C++ Concurrency in Action	2.1 Basic thread management (launching, joining, detaching, waiting in exceptional circumstances, running in the background)	40 to 47
	C++ Concurrency in Action	2.2 Passing arguments to a thread function	47 to 50
	C++ Concurrency in Action	2.3 Transferring ownership of a thread; 2.5 Identifying threads	50 to 53 and 57 to 59
Shared data and data races	C++ Concurrency in Action	3.1 Problems with sharing data between threads (invariants, race conditions)	60 to 63
Mutexes: taking turns	Effective Modern C++	Item 16, Make const member functions thread safe	121 to 127
	C++ Concurrency in Action	3.2.1 to 3.2.3 Mutexes in C++, structuring code to protect data, race conditions inherent in interfaces	64 to 74
	C++ Concurrency in Action	3.2.6 to 3.2.8 std::unique_lock, transferring mutex ownership, locking granularity	82 to 87
Other mutex types and call_once	C++ Concurrency in Action	3.3.1 Protecting shared data during initialization (std::call_once)	88 to 91
	C++ Concurrency in Action	3.3.2 to 3.3.3 Protecting rarely updated data (reader-writer lock); recursive locking	91 to 95
Races hidden in interfaces	C++ Concurrency in Action	3.2.1 to 3.2.3 Mutexes in C++, structuring code to protect data, race conditions inherent in interfaces	64 to 74
	C++ Concurrency in Action	6.1 to 6.2.2 Designing for concurrency, a thread-safe stack and a queue with condition variables	197 to 206
Deadlock	C++ Concurrency in Action	3.2.4 to 3.2.5 Deadlock: the problem, the solution and further guidelines	74 to 82

SECTION	BOOK	PART OF THE BOOK	PDF PAGES
Condition variables	Effective Modern C++	Item 39, Consider void futures for one-shot event communication	280 to 289
	C++ Concurrency in Action	4.1 Waiting for an event: condition variables and a thread-safe queue	96 to 104
	C++ Concurrency in Action	6.1 to 6.2.2 Designing for concurrency, a thread-safe stack and a queue with condition variables	197 to 206
Debugging and common bugs	C++ Concurrency in Action	11.1 Types of concurrency-related bugs, and 11.2.1 reviewing code to locate them	363 to 367